

Detecting Citation Drift in AI-Generated Literature Summaries

Abstract

Large Language Models (LLMs) are increasingly used to generate literature reviews and academic summaries, but they often produce inaccurate or fabricated citations. Such “citation drift” – references that mutate, disappear, or become hallucinated – undermines trust and reproducibility in AI-assisted writing [51†L81-L89] [40†L59-L67]. For example, a recent analysis found that up to 90% of statements from LLM medical Q&A are unsupported by their cited sources [28†L65-L73], and Nature reported “tens of thousands” of 2025 publications may contain invalid AI-generated references [40†L59-L67] [34†L19-L23]. This paper surveys the problem of citation drift in AI-generated summaries, reviews related work on factual consistency and citation verification, and proposes methods to detect drift. We outline new metrics (stability, fabrication rate, drift rate/entropy, willingness-to-cite) and algorithms (retrieval and entailment checks) for identifying mis-citations, leveraging both retrieval-based comparison and LLM-based reasoning [51†L87-L90] [26†L77-L86]. We discuss evaluation on emerging benchmark datasets of AI-generated summaries (e.g. CS-PaperSum, MASSW) and highlight challenges such as reference extraction errors and domain variability [40†L37-L43] [58†L1-L4]. Finally, we identify research gaps and future directions for improving citation fidelity in AI summarization.

Keywords: AI-generated summaries, citation drift, hallucinated references, factual consistency, large language models, literature review, verification.

Introduction

As large language models (LLMs) become ubiquitous in academic writing tools, they are often tasked with generating literature reviews and research summaries. While these models produce fluent prose, their citation behavior can be problematic [51†L81-L89] [34†L19-L23]. Indeed, LLMs frequently fabricate or distort references: for example, Ram (2025) observed that some LLM configurations *fabricate up to 85.6% of citations* in extended interactions [51†L87-L90]. In medicine, Riantawan *et al.* (2025) report that **50-90%** of answers from advanced LLMs are *not fully supported* by their cited sources [28†L65-L73]. A Nature investigation estimates that “tens of thousands” of recent papers contain *AI-generated invalid references* [40†L59-L67] [34†L19-L23]. These findings echo wider concerns: Up to 41% of published papers have reference errors, and LLMs exacerbate “citation hallucination” by inventing plausible but false sources [33†L25-L33] [51†L87-L90].

This paper addresses **citation drift** in AI-generated literature summaries – the phenomenon where references in a summary deviate from the factual content or sources of the original document. *Citation drift* may manifest as misattributed sources, missing citations, or entirely fabricated references. Formally, Ram (2025) defines citation drift as references that “mutate, disappear, or get fabricated during extended LLM interactions” [51†L81-L89]. Though first studied in multi-turn dialogs, similar issues arise in one-shot

summarization: an AI model may summarize a paper or a field and attach citations that do not match the true source material. This undermines reproducibility and trust.

In this paper, we survey the state of research on detecting and quantifying citation drift in LLM-generated summaries. We review related work on citation accuracy, summarization faithfulness, and citation verification. We then propose metrics and detection methods – inspired by *citation stability* measures [51†L87-L90] and fact-checking techniques [32†L123-L132] [26†L77-L86] – to flag mis-citations. We outline how to construct or leverage datasets of summaries with ground-truth sources (e.g. CS-PaperSum [42†L65-L74], MASSW [47†L23-L27]) for evaluation. Finally, we identify challenges (such as imperfect source access and varied citation styles) and gaps (lack of benchmarks) that point to future research directions in this emerging area.

Background and Related Work

Citation Accuracy in AI Writing. The literature on LLM-driven citation emphasizes that citation errors are widespread and increasing. Badalova and Mayr (2026) surveyed existing tools (CheckIfExist, Hallucinator, HalRef, etc.) and concluded that *hallucinated and suspicious references are a growing threat to scholarly communication* [40†L59-L67] [40†L37-L43]. For instance, a Lancet audit found fabricated citations rising in biomedical papers [40†L59-L67]. Moreover, Walters (2024) and others have shown ChatGPT often returns *entirely fabricated or erroneous citations* [40†L69-L77]. Recent conference reports confirm AI-related mis-citations in venues like ICLR and NeurIPS 2026, with ACL 2026 finding over 100 accepted papers containing false references [40†L69-L77]. These analyses underscore the urgency of detecting mis-citations.

Summarization Datasets. To evaluate citation fidelity, one needs datasets of documents and summaries. Traditional summarization corpora (CNN/DailyMail, XSum) lack citations, but scientific summarization datasets exist (e.g. SumPubMed [30†L78-L86], SciTldr). More recently, AI-generated summary datasets have emerged. For example, Liu *et al.* (2025) released **CS-PaperSum**, containing 91,919 computer-science papers with GPT-3.5-generated structured summaries [42†L65-L74]. Zhang *et al.* (2025) introduced **MASSW**, a corpus of 152K CS papers with LLM-extracted “context, methods, outcomes” summaries [47†L23-L27]. These large corpora pair original documents with LLM summaries, providing a testbed for citation analysis. However, most do not explicitly include reference markers or ground-truth citations, so additional annotation or pairing with source text is needed for our task.

Factuality in Summarization. Citation drift is related to the broader issue of faithfulness in text generation. Numerous works have studied hallucinations in summarization and proposed metrics (e.g. ROUGE, BLEU, BERTScore) and more semantic approaches (FactCC, SummaC, QuestEval, Q²) [32†L123-L132]. Such metrics typically evaluate whether summary content is entailed by the source text, but do not directly address the correctness of citation links. Some methods generate questions from summaries and compare to source answers (FEQA [32†L123-L132]). In the clinical domain, Xu *et al.* (2025) built controlled “leave-N-out” datasets to induce summarization hallucinations, and showed that LLM-based detectors (e.g. claim alignment) outperform traditional metrics [32†L123-L132] [32†L139-L144]. While not citation-specific, these ideas (QA-based checking, entailment scoring) can inform citation verification.

Citation Verification Systems. In parallel, the scientific integrity community has developed tools to validate references. Haan (2025) introduces **SemanticCite**, an AI system that verifies citation accuracy via

full-text analysis and classifies each citation as Supported/Partially/Unsupported [26†L77-L86] . RefLens (Lee *et al.*, 2026) is an end-to-end agent framework that extracts evidence spans from original sources and presents an interactive dashboard for citation verification [33†L25-L33] . These systems demonstrate that with sufficient retrieval of source documents, one can *ground* summary claims in evidence. However, they have focused on post-publication verification of written claims, rather than on the generative step. Our work intersects with these efforts by applying evidence-grounded verification to the outputs of summarization models.

Hallucinated References. Agrawal *et al.* (2023) studied hallucinated book/paper references as a “model organism” for LLM hallucination [60†L23-L32] . They found that LMs often *know* when they hallucinate: for fake citations they give inconsistent author lists across queries, whereas real references yield consistent details [60†L23-L32] . This suggests a detection avenue via internal consistency checks. Similarly, Zhao *et al.* (2026) showed that LLM-related hallucinations occur systematically across large corpora [40†L59-L67] . These works highlight that hallucinated references have detectable patterns, motivating explicit detectors.

In sum, prior work has documented the **prevalence** of citation errors [40†L59-L67] [34†L19-L23] and explored components of the problem (citation recommendation accuracy [53†L128-L136] , summarization hallucinations [32†L123-L132] , verification tools [26†L77-L86] [33†L25-L33]). However, **detecting citation drift in AI-generated summaries** remains underexplored. We draw on these literatures to design detection methods tailored to this task.

Metrics and Detection Approach

To quantify citation drift, we adopt and extend metrics introduced by Ram (2025) for multi-turn conversations [51†L87-L90] . Let a summary (or a turn) have a set of cited references C_t . We define:

- **Citation Stability:** The Jaccard overlap between citations across contexts, e.g. $\frac{|C_t \cap C_{t+1}|}{|C_t \cup C_{t+1}|}$. For summarization, one could compare the set of citations in the summary to those in the original paper (if available). Higher stability means fewer dropped or changed citations.
- **Fabrication Rate:** The fraction of citations in the summary that do *not* correspond to any source (i.e., likely invented). For example, if an AI summary cites ten papers but only six appear in the source list, fabrication rate is 0.4.
- **Drift Rate:** The normalized symmetric difference of citation sets (i.e. proportion changed). In multi-turn, this measured how many citations changed per turn; for a one-shot summary, drift rate can measure divergence from the known source citations.
- **Drift Entropy:** An information-theoretic measure of unpredictability in reference changes (originally defined over conversational history) [51†L87-L90] . In our case, it could capture the variability of citation presence when re-generating a summary under different seeds.
- **Willingness-to-Cite:** A binary metric indicating if any citations are present at all. Some AI summaries may omit references entirely; this flag helps quantify the model’s propensity to cite [51†L87-L90] .

These metrics provide a framework for evaluation and can be computed once references are extracted. However, the core task is **detecting which specific citations are drifted or false**. We propose the following detection pipeline:

1. **Reference Extraction:** Parse the AI-generated summary for reference markers (e.g. numerical indices, DOIs, titles). Tools like GROBID or neural parsers can convert bibliographic entries in text into structured data. Errors at this stage (failed extraction, misrecognized style) can degrade detection performance.
2. **Source Retrieval:** For each cited reference, use metadata (DOI, title, authors) to retrieve the original document or authoritative summary (via CrossRef, Semantic Scholar, arXiv, etc.). If the summary originates from a known corpus (e.g. CS-PaperSum, MASSW), this step may involve matching to the corpus entry.
3. **Content Alignment:** Using the retrieved source text, identify the most relevant passages corresponding to each citation. This can be done via simple methods (keyword search for unique terms from the citation context) or dense retrieval (embedding similarity). For example, if the summary says “[Smith et al. 2020] show X”, we locate where Smith2020 appears or where related content occurs in the source.
4. **Factual Verification:** For each summary sentence or claim that cites a reference, apply a fact-checking model to assess support. Options include:
5. **Natural Language Inference (NLI):** Use an entailment model (e.g. SciBERT-based or LLM prompt) to score whether the summary claim is entailed by the aligned source sentence.
6. **Question Answering:** Formulate the claim as a question and see if the answer appears in the source snippet.
7. **LLM Probing:** As Agrawal *et al.* (2023) suggest [60†L23-L32], one can directly ask the LLM about the reference. For example, “List the authors of [Smith 2020].” Inconsistencies indicate a likely hallucination. Similarly, one can prompt the LLM to verify whether the cited paper contains a given fact.
8. **Aggregate Decision:** Combine the evidence to label each citation as *correctly supported*, *unsupported*, or *hallucinated*. For instance, if the NLI score is low or the LLM cannot recall the reference details correctly, flag the citation. The classification scheme used by SemanticCite (Supported/Partially/Unsupported) [26†L77-L86] can be adapted: “Supported” means the citation aligns well with the content, while “Unsupported” indicates drift or fabrication.
9. **Metric Computation:** With the labels, compute overall drift metrics (fabrication rate, etc.) and an error rate (fraction of citations mis-cited). This yields a performance measure for the detection method.

Importantly, such an approach must tolerate cases where the source text may not contain a direct quote of the summary claim. For multi-document summaries (e.g. summarizing a field), a cited reference should be

relevant, not necessarily entailed verbatim by the summary sentence. In these cases, semantic relevance and topical alignment may suffice to consider a citation plausible.

Current Approaches and Tools

Several recent efforts align with the above strategy. **SemanticCite** (Haan, 2025) is an AI system that automates citation verification by retrieving source PDFs and analyzing textual evidence [26†L77-L86]. It classifies each citation into support categories and reports high-quality dataset creation (1,000+ annotated citation claims) [26†L77-L86]. Adapting SemanticCite to summaries would involve feeding it the summary claims and checking against sources.

Reflens (Lee *et al.*, 2026) is an end-to-end verification framework that parses PDF references and displays evidence-based reports [33†L25-L33]. It exemplifies the retrieval-and-extraction approach: the system downloads referenced PDFs and finds verbatim evidence for each citation. Incorporating such a pipeline, an AI summarizer could highlight evidence spans corresponding to each summary sentence, assisting in human evaluation of drift.

SourceCheckup (Riantawan *et al.*, 2025) implements an agent-based pipeline for medical QA. It automatically generates questions, obtains LLM answers with cited sources, and then scores how well those sources support the answers [28†L65-L73]. Their finding – that even GPT-4 with Web search still has ~30–40% unsupported statements [28†L65-L73] – underscores the difficulty. While SourceCheckup targets open-domain QA, its methodology (parsing statements and URLs) is analogous to our summary citation verification.

Hallucinator tools: Some open-source tools already scan papers for fake references. Badalova and Mayr (2026) evaluated tools like *HalluCiteChecker* and *Hallucinatory Reference Finder*. They note these tools often rely on cross-referencing metadata or simple name matching [40†L37-L43], so they can catch obvious fakes (nonexistent titles) but fail on subtle drifts. For example, an LLM might cite a real paper but attribute it to the wrong claim; such “suspicious” cases require semantic checks beyond metadata (a limitation noted in existing systems [40†L37-L43]).

LLM self-verification: Agrawal *et al.* (2023) proposed directly querying the LLM about its own references [60†L23-L32]. In their experiments, asking a model for details (authors, year) of a cited paper helped discriminate real vs. fake references, because inconsistent answers signaled hallucinations [60†L23-L32]. One could integrate this by prompting the same model: e.g. after generating a summary with references, ask it to verify each citation. However, this relies on the model’s own internal knowledge and may suffer from overconfidence.

Benchmark datasets: To date, there is no standard benchmark explicitly for citation drift detection. However, the emerging large-scale summary datasets (CS-PaperSum [42†L65-L74], MASSW [47†L23-L27]) provide synthetic testbeds. Researchers could take an AI-generated summary and artificially inject citation errors or compare LLM summaries to human abstracts. Additionally, Haan’s 1,000-citation dataset [26†L77-L86] provides aligned summary-claim and citation pairs (with labels) that could be adapted for single-document summarization cases.

Challenges and Limitations

Detecting citation drift faces several inherent challenges:

- **Reference Extraction Errors:** Automatically parsing reference text or mapping in-text citations to database entries is error-prone. Inconsistent citation styles or OCR noise (if reading PDFs) can cause mismatches [40†L37-L43] . Many tools struggle when metadata is incomplete or nonstandard.
- **Lack of Explicit Context:** Unlike fact-checking where the claim and evidence can be co-located, citations may refer to broad content. A citation supporting a broad claim (“As shown in Smith et al. 2021...”) may not have a single keyword overlap. Aligning summary claims with source text may require elaborate retrieval or human-level understanding.
- **Multi-Document Summaries:** In a literature review setting, summaries may synthesize multiple papers. A citation might be relevant but not literally present in the combined summary text. Deciding support in this fuzzy scenario is hard; our methods must allow some semantic flexibility.
- **Model Non-Determinism:** As Ram (2025) noted, LLM citations can vary across runs [51†L87-L90] . A citation that appears in one generation may vanish in another. This non-determinism (drift entropy) complicates evaluation: how to label a citation as “wrong” if it’s not consistently produced? A detection system may need multiple samples or focus on ones consistently output.
- **Evaluation Data:** There is no large labeled corpus of AI summaries with annotated true/false citations. Creating such a dataset is labor-intensive. Without it, benchmarking detection models relies on proxy tasks or synthetic data (e.g. hiding certain references, as done by Xu *et al.* [32†L49-L58] in the medical domain). The lack of benchmarks hinders progress.
- **Cross-Domain Variation:** Citations have different conventions in different fields (e.g. computer science vs. medicine vs. social sciences). A one-size-fits-all detector may misinterpret domain-specific citation practices. For example, summarizing code repositories or mathematical proofs introduces new citation forms.
- **LLM Limitations:** Any detection method that uses an LLM to verify citations is subject to the model’s own hallucinations or biases. An LLM might confidently assert a false citation is real, or fail to recognize a true citation under paraphrase.

Given these challenges, current tools can only provide *alerts*, not definitive judgments [40†L37-L43] . Detection systems must combine multiple signals (metadata, retrieval, semantic checks) and operate in a probabilistic fashion.

Research Gaps and Future Directions

Standardized Benchmarks: A key gap is the absence of shared benchmarks for citation drift. We envision creating a dataset of AI-generated summaries with annotated citations (correct vs. drifted) across domains. This could involve human annotators verifying each citation in a corpus like CS-PaperSum against the original paper or other literature. Such a benchmark would allow objective comparison of drift detectors.

Advanced Semantic Checking: Future work should improve the *semantic* analysis of citations. For instance, developing an entailment model fine-tuned on scientific text claims could better judge whether a citation truly supports a claim. Machine reading comprehension on scientific documents (beyond simple QA) is needed to bridge the gap between citation metadata and content.

Interactive and Multi-Agent Approaches: Multi-agent frameworks like RefLens suggest a path: one agent retrieves sources, another extracts evidence, a third evaluates support, etc. Such modular systems could be tailored to summarization pipelines. Research into agents specifically designed for literature reviews (e.g. an “Information Retrieval Agent” that fetches papers vs. a “Verification Agent” that checks each citation) is a promising direction.

Model-Aware Generation: Instead of post-hoc detection, generation-time mitigation is important. For example, a summarization model could be trained to copy reference lists from sources (“source copying”) or to cite only items it has evidence for. Reinforcement learning with human feedback (RLHF) could penalize hallucinated references. Early research (e.g. “citation-enhanced generation” by Li *et al.*, 2024) hints at this idea. Integrating citation-checks into the generation loop remains an open challenge.

Cross-Document Summarization: Citation drift is particularly thorny when summarizing across multiple documents (systematic reviews, thematic analyses). Future research should study detection in this setting: e.g. distinguishing when a citation in a multi-document summary belongs to which source. Knowledge graphs or citation networks might help constrain possible references.

User Interfaces and Tools: Beyond research algorithms, there is a need for user-friendly tools. For instance, word-processing plugins that flag suspect citations in a draft summary, or citation “linters” for AI outputs. Incorporating detection into writing environments (e.g. Overleaf, SciNote) would raise awareness and improve scientific writing hygiene.

Longitudinal and Empirical Studies: As AI adoption grows, empirical studies on how citation drift affects scientific discourse are important. For example, measuring how often AI-written papers are retracted for false citations, or surveying researchers on trust in AI-generated references. Such data could motivate stricter standards for AI-assisted writing.

In summary, detecting citation drift in AI summaries is a multi-faceted problem bridging NLP, IR, and scholarly communication. While initial metrics and tools exist [51†L87-L90] [26†L77-L86], much work remains to build robust, generalizable detection systems.

Conclusion

Citation drift – the corruption of references in AI-generated summaries – poses a serious risk to research integrity. We have outlined the dimensions of this problem, drawing on evidence from recent studies that LLMs hallucinate references at alarming rates [51†L81-L89] [40†L59-L67]. To tackle this, we propose a framework of detection: define metrics (stability, fabrication rate, etc.), employ retrieval and semantic verification of citations [51†L87-L90] [26†L77-L86], and leverage large summary datasets for evaluation [42†L65-L74] [47†L23-L27]. Our analysis shows that existing tools can provide preliminary warnings (via metadata checks [40†L37-L43]) but often fail to catch subtle drift. Ultimately, mitigating citation drift will require combining these detection methods with improvements in summary generation. We anticipate that

robust benchmarks and interdisciplinary research – drawing on fact-checking, information retrieval, and publishing ethics – will be crucial to ensure AI-assisted literature reviews remain reliable.

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